$$H = \begin{pmatrix} E+m & ke^{-i\theta} \\ ke^{i\theta} & E-m \end{pmatrix}$$

P: a given projector

eff. H 会是 P 的空间上的算符,

ev: ϵ_α

es: $P|\psi_\alpha\rangle$

↳ 本来的 H 的 es

e.g. 考虑 $P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

wave operator Ω s.t. $|\psi_\alpha\rangle = \Omega P |\psi_\alpha\rangle$

$$\Longrightarrow H_{eff} = PH\Omega P \longrightarrow H_{eff}(P|\psi_\alpha\rangle) = PH\Omega PP|\psi_\alpha\rangle$$

$$= PH\Omega P|\psi_\alpha\rangle = PH|\psi_\alpha\rangle$$

$$= \epsilon_\alpha (P|\psi_\alpha\rangle)$$

$$(1-P)H|\psi_\alpha\rangle = (1-P)\epsilon_\alpha|\psi_\alpha\rangle$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} H|\psi_\alpha\rangle = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \epsilon_\alpha |\psi_\alpha\rangle$$

$$\begin{pmatrix} 0 & 0 \\ ke^{i\theta} & E-m \end{pmatrix} |\psi_\alpha\rangle = \epsilon_\alpha \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} |\psi_\alpha\rangle$$

$$\begin{pmatrix} 0 & 0 \\ ke^{i\theta} & E-m \end{pmatrix} \begin{pmatrix} u_\alpha \\ d_\alpha \end{pmatrix} = \epsilon_\alpha \begin{pmatrix} 0 \\ d_\alpha \end{pmatrix}$$

i.e. $\begin{pmatrix} 0 \\ ke^{i\theta}u_\alpha + d_\alpha(E-m) \end{pmatrix} = \begin{pmatrix} 0 \\ d_\alpha \epsilon_\alpha \end{pmatrix}$

i.e. $\epsilon_\alpha d_\alpha = ke^{i\theta}u_\alpha + (E-m)d_\alpha$

i.e. $d_\alpha = \dfrac{ke^{i\theta}}{\epsilon_\alpha - E + m} u_\alpha$

$$\Longrightarrow \begin{pmatrix} u_\alpha \\ d_\alpha \end{pmatrix} \longmapsto \begin{pmatrix} u_\alpha \\ d_\alpha \end{pmatrix}$$

$$\Omega : P|\psi_\alpha\rangle \longrightarrow |\psi_\alpha\rangle$$

$$\Omega P = \begin{pmatrix} 1 & 0 \\ \dfrac{ke^{i\theta}}{\epsilon_\alpha - E + m} & 0 \end{pmatrix} \quad \checkmark$$

$$\Omega P \begin{pmatrix} u_\alpha \\ d_\alpha \end{pmatrix} = \begin{pmatrix} u_\alpha \\ d_\alpha \end{pmatrix}$$

$$\Longrightarrow H_{eff} = PH\Omega P = \begin{pmatrix} E+m & ke^{-i\theta} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \dfrac{ke^{i\theta}}{\epsilon_\alpha - E + m} & 0 \end{pmatrix} = \begin{pmatrix} E+m+\dfrac{k^2}{\epsilon_\alpha - E + m} & 0 \\ 0 & 0 \end{pmatrix}$$

$$H_{eff} \begin{pmatrix} u_\alpha \\ 0 \end{pmatrix} = \epsilon_\alpha \begin{pmatrix} u_\alpha \\ 0 \end{pmatrix}, \text{ i.e. } \left(E+m+\frac{k^2}{\epsilon_\alpha - E + m}\right) u_\alpha = \epsilon_\alpha u_\alpha$$

$$\Longrightarrow \epsilon_\alpha = E + m + \frac{k^2}{\epsilon_\alpha - E + m}$$

$$\epsilon_\alpha^2 - E\epsilon_\alpha \cancel{+m\epsilon_\alpha} = E\epsilon_\alpha - E^2 \cancel{+Em} \cancel{+m\epsilon_\alpha} \cancel{-mE} + m^2 + k^2$$

$$\epsilon_\alpha^2 - 2E\epsilon_\alpha + (E^2 - m^2 - k^2) = 0$$

$$\epsilon_\alpha = \frac{1}{2}\left(2E \pm \sqrt{4E^2 - 4E^2 + 4m^2 + 4k^2}\right) = E \pm \sqrt{m^2 + k^2}$$